CATALYST. BY MRS. RANA · GRADE 9 · SCIENCE · TERM 1

Term Mock Exam

Usual ministry format · 30 questions · 100 marks · 120 minutes

Name _____ Class _____ Score _____ / 100

PART 1 · MULTIPLE CHOICE · 25 × 2 MARKS

1 A scientist records which birds visit a pond and how water levels change each month. What is the scientist studying?عالم يسجل الطيور ومستوى الماء... — 2 marks

A Ecology
B Chemistry
C Geology
D Astronomy

2 Which statement best compares biotic and abiotic factors?قارني. — 2 marks

A Abiotic factors can reproduce; biotic factors cannot.
B Biotic factors are living parts of an ecosystem; abiotic factors are nonliving parts.
C Biotic factors are always animals; abiotic factors are always plants.
D Both are living parts of an ecosystem.

3 A student lists: oryx, ghaf trees, sand, sunlight and low rainfall in one desert area. Which level does the list describe?قائمة فيها كائنات وعوامل غير حية... — 2 marks

A Population
B Biological community
C Ecosystem
D Organism

4 A woodpecker lives in an oak tree and eats insects under its bark. Which statement is correct?نقار الخشب... — 2 marks

A It has a habitat but no niche.
B Habitat and niche both mean the food it eats.
C The oak tree is its niche; eating insects is its habitat.
D The oak tree is its habitat; eating insects under the bark is part of its niche.

5 What is the primary role of decomposers?؟الدور الأساسي للمحللات — 2 marks

A To produce oxygen
B To hunt herbivores
C To capture sunlight
D To break down organic compounds and make nutrients available to producers

6 A student draws: snake → mouse → grasshopper → grass, meaning the arrows show energy flow. Which change corrects the model?طالبة رسمت السلسلة بالعكس... — 2 marks

A Remove the grass.
B Reverse all of the arrows.
C Replace the mouse with a producer.
D Put the snake between the grass and the grasshopper.

7 Why must matter cycle through the biosphere?؟ليه المادة لازم تدور — 2 marks

A Matter is not created or destroyed, so new life is built from existing atoms
B Plants destroy matter
C The Sun makes new matter every day
D Energy is recycled with matter

8 Carbon enters a long-term cycle when organic matter is buried underground and converted to...الكربون يدخل دورة طويلة لما يتحول إلى... — 2 marks

A oxygen
B water vapor
C fossil fuels such as coal, oil or gas
D ammonium

9 High nitrogen content in runoff from fertilizer can cause...النيتروجين في جريان السماد يسبب... — 2 marks

A more nitrogen fixation
B fewer algae
C algae blooms
D acid-free lakes

10 When organisms die or produce waste, which organisms return phosphorus to the soil?مين يرجع الفوسفور للتربة؟ **2 marks**

- **A** Producers
- **B** Decomposers
- **C** Herbivores
- **D** Carnivores

11 Why does genetic diversity help a population survive?ليه التنوع الجيني يساعد على البقاء؟ **2 marks**

- **A** Some individuals may survive changing conditions or an outbreak of disease
- **B** It removes predators
- **C** It stops reproduction
- **D** All individuals become identical

12 New York City chose to protect its watersheds instead of building a filtration system because...ليه نيويورك حمت مستجمعات المياه؟ **2 marks**

- **A** the watersheds were in another country
- **B** people prefer bottled water
- **C** filtration was not possible
- **D** preserving the watershed (≈$1.5 billion) was cheaper than technology (>$6 billion)

13 Some scientists estimate the current extinction rate is about how many times the normal background rate?كم ضعف المعدل الطبيعي؟ **2 marks**

- **A** 100,000 times
- **B** 10 times
- **C** 1000 times
- **D** 2 times

14 How is today's high rate of extinction different from past mass extinctions?إيه الفرق عن الانقراضات السابقة؟ **2 marks**

- **A** Humans are changing conditions faster than new traits can evolve
- **B** It does not affect plants
- **C** Biodiversity recovers in a few years
- **D** It is caused only by asteroids

15 Why do invasive species often reproduce in large numbers in a new area?ليه الأنواع الغازية تتكاثر بكثرة؟ **2 marks**

- **A** They don't need food
- **B** Native species help them
- **C** They are always larger
- **D** There is a lack of predators, parasites and competition that controlled them at home

16 Natural resource consumption per person is much higher in...الاستهلاك للفرد أعلى في... **2 marks**

- **A** countries that rely on agriculture
- **B** developed countries
- **C** every country equally
- **D** developing countries

17 Approximately what fraction of all plant and animal species are found in hotspots?نسبة الأنواع في النقاط الساخنة؟ **2 marks**

- **A** About one tenth
- **B** About half
- **C** About 1%
- **D** Almost all

18 How do some plants remove toxic metals such as zinc, lead and nickel from soil?إزاي النبات يشيل المعادن السامة؟ **2 marks**

- **A** They push metals deeper underground
- **B** They turn metals into oxygen
- **C** They store the metals in their tissues, then the plants are harvested
- **D** They eat the metals and release them as gas

19 In J.J. Thomson's model (1904), the atom was...نموذج طومسون... **2 marks**

A a nucleus surrounded by an electron cloud
B a solid ball with no electrons
C a nucleus with electrons orbiting like planets
D electrons embedded in a ball of positive charge

20 Which statement about the scientific methods is correct?عبارة صحيحة؟ **2 marks**

A Scientists may add steps, repeat steps or skip steps
B Every scientist must follow the same rigid steps
C Conclusions always come first
D There is only one step

21 Why might a scientist use a computer simulation instead of only a real experiment?ليه المحاكاة؟ **2 marks**

A Simulations need no reasoning
B Simulations are always more accurate than real data
C It can safely mimic processes that are too large, small, slow or dangerous to test
D Real experiments are never needed

22 Which question would an ecologist MOST likely investigate?أي سؤال يدرسه عالم البيئة؟ **2 marks**

A What minerals make up granite?
B How does the amount of rainfall affect the number of frogs in a wetland?
C How fast does a car travel?
D What is the boiling point of water?

23 At 23°C, steelhead trout are in the...سمك التراوت في 23 °C عند... **2 marks**

A optimum range
B zone of intolerance
C lower limit
D zone of physiological stress

24 Which example is a biological community?مثال للمجتمع؟ **2 marks**

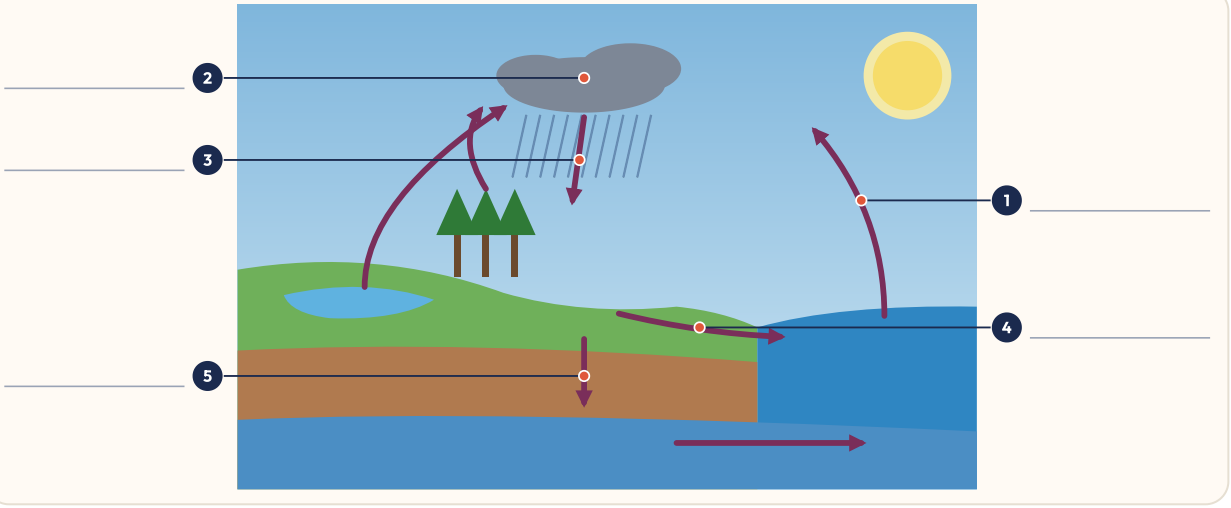
A Fish, coral, turtles and sea plants on one reef
B A school of the same fish species
C Only the sea water and sunlight on a reef
D One clownfish

25 One bird eats insects on leaves while another uses bark for its nest in the same tree. Why does this increase their chance of survival?ليه ده يزود فرص البقاء؟ **2 marks**

A They become one species
B They compete more strongly
C The tree grows faster
D They use the resources in different ways

PART 2 · WRITTEN ANSWERS · 5 × 10 MARKS

26 The diagram shows the water cycle. a) Name the processes labeled 1–5. (5 marks) b) About 10% of water vapor comes from plants. Name this process and explain how clouds form. (5 marks) **10 marks**



27 A food chain is grass → grasshopper → mouse → snake. The grass stores 20,000 kJ of energy. a) Calculate the energy available to the grasshopper, the mouse and the snake. Show your working. (6 marks) b) Explain why only about 10% of the energy passes to the next trophic level. (4 marks) **10 marks**

28 Steelhead trout survive in water from 9°C to 25°C, and the most trout are found between 13°C and 21°C. a) State the range of tolerance and the optimum range. (4 marks) b) Name the zone a trout is in at 23°C and describe what happens to it. (4 marks) c) Is water temperature a biotic or abiotic limiting factor? (2 marks) **10 marks**

29 A pesticide enters a lake at 0.000003 ppm and reaches 25 ppm in fish-eating birds. a) Name this process and define it. (4 marks) b) Explain why the birds have the highest concentration. (4 marks) c) Give one effect of DDT on fish-eating birds. (2 marks) **10 marks**

30 A student investigates how the amount of fertilizer affects the height of bean plants. a) Identify the independent and dependent variables. (4 marks) b) Name two constants. (2 marks) c) Describe a suitable control. (2 marks) d) What should the student do if the results do not support the hypothesis? (2 marks) **10 marks**